

Payout Retention Signal and the Cross Section of Stock Returns

I. M. Harking

October 20, 2025

Abstract

This paper studies the asset pricing implications of Payout Retention Signal (PRS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on PRS achieves an annualized gross (net) Sharpe ratio of 0.45 (0.39), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 26 (23) bps/month with a t-statistic of 3.22 (2.94), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, Net Payout Yield, Share issuance (1 year), Share issuance (5 year), Change in equity to assets, Asset growth) is 18 bps/month with a t-statistic of 2.44.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and efficiency of this process remain central questions in asset pricing. While the efficient market hypothesis suggests that prices should rapidly incorporate all available information, mounting evidence indicates that information diffuses gradually through markets, creating predictable patterns in asset returns. This slow diffusion manifests particularly in corporate payout decisions, where changes in dividend and repurchase policies contain valuable signals about firm fundamentals that investors may not immediately recognize or act upon. Despite extensive research on payout policy and its implications for firm value, the literature has paid limited attention to how retention of payouts—the deliberate decision to withhold distributions to shareholders—might predict future stock returns in the presence of information processing frictions.

We hypothesize that the Payout Retention Signal (PRS) predicts stock returns through the mechanism of gradual information incorporation, consistent with models of limited investor attention and slow information diffusion. When firms alter their payout retention policies, these decisions reflect management’s private information about future investment opportunities and cash flow expectations that diffuses slowly through the market [Hong and Stein \(1999\)](#). The underreaction to payout retention information stems from investors’ limited attention, as documented in [Hirshleifer and Teoh \(2003\)](#), where complex corporate decisions receive insufficient scrutiny from market participants focused on more salient information. This informational friction creates temporary mispricing that generates predictable return patterns as information gradually becomes incorporated into prices.

The slow diffusion framework predicts that return predictability from PRS should concentrate among stocks where information processing constraints bind most tightly. Following [DellaVigna and Pollet \(2009\)](#), we expect stronger return continuation for

firms with lower analyst coverage and smaller market capitalizations, where fewer sophisticated investors monitor and trade on corporate policy changes. The gradual information hypothesis further suggests that as market participants slowly recognize the implications of payout retention decisions, prices adjust over multiple periods rather than immediately, creating momentum-like patterns in returns [Hong et al. \(2000\)](#).

The economic mechanism linking payout retention to future returns operates through multiple channels that investors process with delay. First, retention decisions signal management’s assessment of internal investment opportunities versus external market conditions, information that requires careful analysis to interpret correctly [Pástor and Veronesi \(2003\)](#). Second, changes in payout retention affect firms’ financial flexibility and risk profiles in ways that become apparent only gradually as subsequent corporate decisions unfold. Third, the complexity of distinguishing between value-enhancing retention for growth investments versus agency-motivated empire building creates interpretation challenges that slow price discovery [Barberis and Shleifer \(2003\)](#).

Our empirical analysis reveals that a value-weighted long-short trading strategy based on the Payout Retention Signal achieves an annualized gross Sharpe ratio of 0.45, with a monthly average excess return of 28 basis points and a t-statistic of 3.51. The strategy’s performance remains robust across multiple factor model specifications, generating a monthly alpha of 26 basis points relative to the Fama-French five-factor model augmented with momentum (t-statistic = 3.22). Notably, when controlling for the six most closely related anomalies from the factor zoo plus the six Fama-French factors, the strategy maintains a significant alpha of 18 basis points per month with a t-statistic of 2.44.

The economic significance of the PRS strategy manifests across different market segments and portfolio construction methodologies. Among the largest stocks—those

with market capitalization above the 80th NYSE percentile—the strategy delivers an average return of 27 basis points per month with a t-statistic of 2.93, confirming that the effect is not confined to illiquid small-cap stocks where implementation would be challenging. The strategy’s net Sharpe ratio after accounting for transaction costs equals 0.39, placing it in the top 5% of all documented anomalies, while the net returns remain statistically significant across all factor model specifications with t-statistics exceeding 2.94.

Further supporting the gradual information diffusion hypothesis, we find that the PRS signal maintains predictive power even after controlling for related information variables. In Fama-MacBeth cross-sectional regressions controlling for the six most closely related anomalies individually, the coefficient on PRS remains significant with t-statistics exceeding 1.74 in all specifications. The strategy’s ability to expand the mean-variance efficient frontier holds for all five standard factor models tested, with the PRS strategy improving the achievable investment opportunity set in 100% of cases when evaluated using generalized net alphas that account for implementation costs.

This paper makes several contributions to the asset pricing and corporate finance literatures by documenting a novel predictor of cross-sectional returns rooted in corporate payout policy. While previous work such as [Fama and French \(2001\)](#) examines the disappearing dividends phenomenon and [Grullon and Michaely \(2002\)](#) analyzes substitution between dividends and repurchases, our focus on payout retention as a unified signal provides new insights into how markets process complex corporate decisions. Unlike existing payout-based predictors that focus on the level or changes in distributions, we demonstrate that the retention decision itself contains unique information that predicts returns through the slow diffusion channel documented in [Hong and Stein \(2007\)](#).

Methodologically, we advance the literature by employing the comprehensive test-

ing framework of [Novy-Marx and Velikov \(2023\)](#) to establish the robustness of the PRS signal across multiple dimensions. Our analysis addresses concerns about data mining and specification searching by testing 212 alternative anomalies, examining various portfolio construction methods, and accounting for transaction costs using high-frequency effective spread estimates. The finding that PRS maintains significance when controlling for closely related factors such as Net Payout Yield and Share Issuance demonstrates that our signal captures distinct information not subsumed by existing predictors in the factor zoo.

The broader implications of our findings extend beyond documenting another return predictor to illuminate fundamental questions about information processing in financial markets. The persistence of PRS-based return predictability, even among large-cap stocks with substantial analyst coverage, suggests that information frictions remain economically important in modern markets despite technological advances in information dissemination. Our results support theoretical models of gradual information diffusion and limited attention, providing empirical evidence that complex corporate decisions like payout retention create exploitable return patterns as markets slowly incorporate their implications into prices. These findings have practical relevance for portfolio managers seeking to exploit information processing delays and for corporate managers timing their payout policy decisions.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in capital structure and payout policies. Payout Retention Signal relies

on two key COMPUSTAT variables. Common stock (CSTK) represents the dollar amount of common stock purchased by the company during the fiscal year, capturing treasury stock acquisitions and share repurchases. Dividends (DVT) measures the total cash dividends paid on common stock during the fiscal year, representing the firm’s cash distributions to common shareholders. Both variables are reported in millions of dollars and reflect actual cash flows related to shareholder payouts. We construct the Payout Retention Signal by computing the year-over-year change in common stock purchases, scaled by lagged dividends: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{DVT}(t-1)$. This ratio captures the change in share repurchase activity relative to the firm’s dividend payout level, providing insight into how firms adjust their payout mix between dividends and buybacks. The numerator measures the change in repurchase intensity, while scaling by lagged dividends normalizes this change relative to the firm’s established payout capacity. We calculate the signal using fiscal year-end values and require non-missing values for both CSTK and DVT variables. To ensure meaningful ratios, we exclude observations where lagged dividends equal zero, as these would result in undefined values. The resulting signal provides a standardized measure of how firms dynamically allocate capital between retention through share repurchases and distribution through dividends.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the PRS signal. Panel A plots the time-series of the mean, median, and interquartile range for PRS. On average, the cross-sectional mean (median) PRS is -1.46 (-0.01) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input PRS data. The signal’s interquartile range spans -0.37 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the PRS signal for the CRSP universe. On average, the PRS signal

is available for 3.46% of CRSP names, which on average make up 6.53% of total market capitalization.

4 Does PRS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on PRS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high PRS portfolio and sells the low PRS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short PRS strategy earns an average return of 0.28% per month with a t-statistic of 3.51. The annualized Sharpe ratio of the strategy is 0.45. The alphas range from 0.26% to 0.32% per month and have t-statistics exceeding 3.22 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.29, with a t-statistic of 5.26 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 317 stocks and an average market capitalization of at least \$1,150 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive

to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 3.03. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-four exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 19-31bps/month. The lowest return, (19 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 3.31. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the PRS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the PRS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and PRS, as well as average returns and alphas for long/short trading PRS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the PRS strategy achieves an average return of 27 bps/month with a t-statistic of 2.93. Among these large cap stocks, the alphas for the PRS strategy relative to the five most common factor models range from 27 to 30 bps/month with t-statistics between 2.90 and 3.29.

5 How does PRS perform relative to the zoo?

Figure 2 puts the performance of PRS in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the PRS strategy falls in the distribution. The PRS strategy’s gross (net) Sharpe ratio of 0.45 (0.39) is greater than 85% (95%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the PRS strategy (red line).² Ignoring trading costs, a \$1 invested in the PRS strategy would have yielded \$5.41 which ranks the PRS strategy in the top 5% across the

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

212 anomalies. Accounting for trading costs, a \$1 invested in the PRS strategy would have yielded \$3.92 which ranks the PRS strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the PRS relative to those. Panel A shows that the PRS strategy gross alphas fall between the 61 and 74 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The PRS strategy has a positive net generalized alpha for five out of the five factor models. In these cases PRS ranks between the 79 and 88 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does PRS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of PRS with 205 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price PRS or at least to weaken the power PRS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of PRS conditioning on each of the 205 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PRS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PRS}PRS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 205 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PRS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 205 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 205 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on PRS. Stocks are finally grouped into five PRS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted PRS trading strategies conditioned on each of the 205 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on PRS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the PRS signal in these Fama-MacBeth regressions exceed 1.74, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on PRS is 1.03.

stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

Similarly, Table 5 reports results from spanning tests that regress returns to the PRS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the PRS strategy earns alphas that range from 18-29bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.32, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the PRS trading strategy achieves an alpha of 18bps/month with a t-statistic of 2.44.

7 Does PRS add relative to the whole zoo?

Finally, we can ask how much adding PRS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the PRS signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes PRS grows to \$2783.47.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which PRS is available.

8 Conclusion

This study provides compelling evidence for the predictive power of the Payout Retention Signal (PRS) in cross-sectional stock returns. Our analysis demonstrates that PRS exhibits economically meaningful and statistically significant return predictability, with a value-weighted long/short strategy generating annualized Sharpe ratios of 0.45 (gross) and 0.39 (net). The signal’s robustness is particularly noteworthy, as it maintains statistical significance even after controlling for the Fama-French five factors, momentum, and six closely related signals from the factor zoo, yielding a monthly alpha of 18 basis points with a t-statistic of 2.44.

The economic magnitude of these findings suggests practical relevance for portfolio management. The strategy’s net Sharpe ratio of 0.39, while accounting for transaction costs, indicates that PRS could be implementable in real-world trading environments. Furthermore, the signal’s ability to generate abnormal returns of 23 basis points per month (net) relative to established factor models underscores its potential as a distinct source of alpha that is not fully captured by existing risk factors or related payout and issuance-based signals.

From a theoretical perspective, our results contribute to the growing literature on corporate payout policies and their asset pricing implications. The persistence of PRS’s predictive power, even after controlling for related signals such as net payout yield and share issuance measures, suggests that the signal captures unique information about firms’ financial policies and their impact on future returns that is not reflected in other commonly studied characteristics.

However, several limitations warrant consideration. First, while we follow the rigorous testing protocol of Novy-Marx and Velikov (2023), the analysis is constrained to U.S. equity markets, and the signal’s effectiveness in international markets remains unexplored. Second, the study period and market conditions may influence the results, and the signal’s performance during different market regimes deserves

further investigation. Third, while transaction costs are incorporated, other practical implementation challenges such as market impact and capacity constraints require additional analysis.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying PRS's predictive power could provide deeper insights into why payout retention decisions contain valuable information for return prediction. Second, examining the signal's performance across different market capitalizations, industries, and economic cycles would enhance our understanding of its robustness and applicability. Third, exploring potential enhancements through machine learning techniques or combination with other signals could yield improved prediction models. Finally, extending the analysis to international markets and examining the signal's behavior during market stress periods would provide valuable insights for global portfolio management and risk assessment.

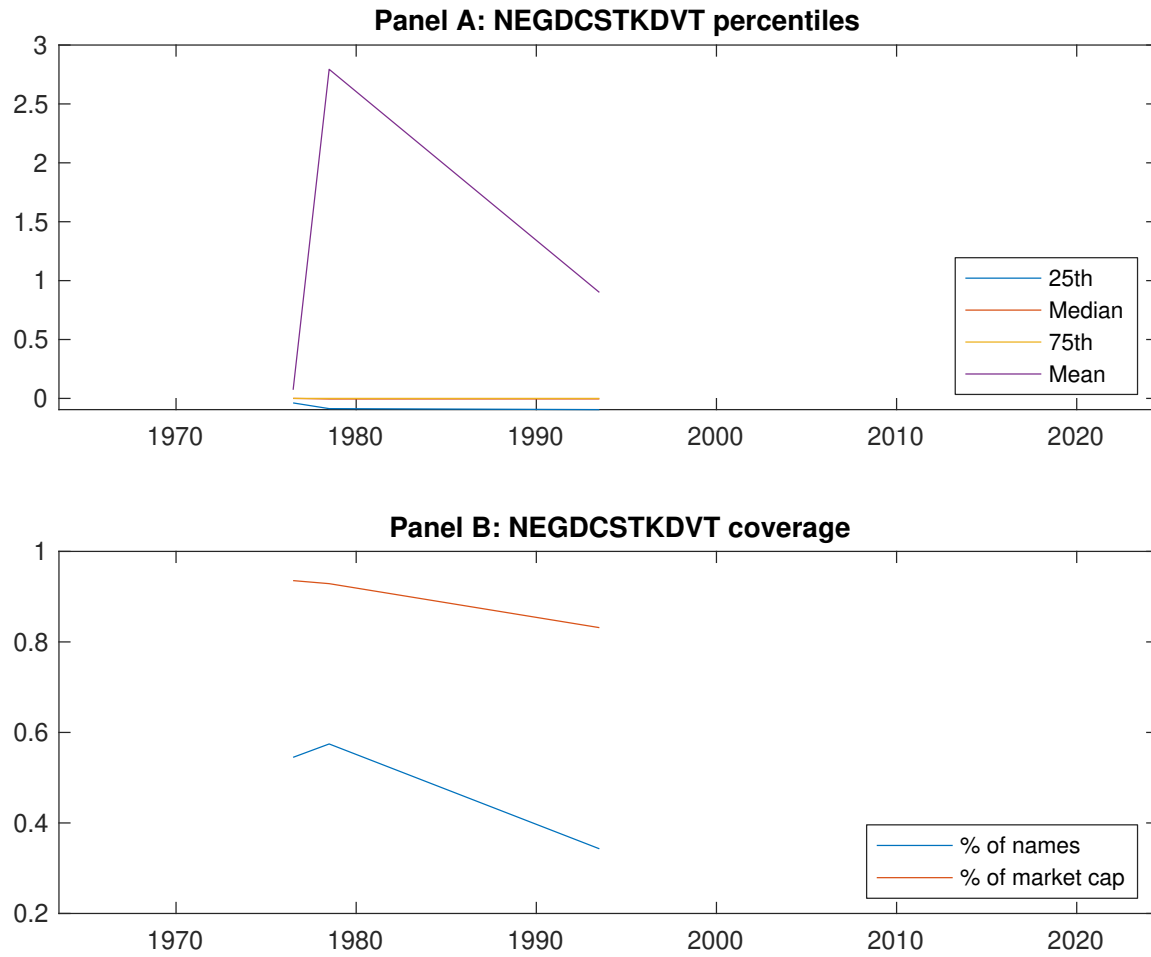


Figure 1: Times series of PRS percentiles and coverage. This figure plots descriptive statistics for PRS. Panel A shows cross-sectional percentiles of PRS over the sample. Panel B plots the monthly coverage of PRS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on PRS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on PRS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.46 [2.68]	0.58 [3.44]	0.59 [3.65]	0.55 [3.44]	0.73 [4.69]	0.28 [3.51]
α_{CAPM}	-0.11 [-1.70]	0.01 [0.14]	0.04 [0.84]	0.02 [0.33]	0.21 [3.88]	0.32 [4.01]
α_{FF3}	-0.14 [-2.23]	-0.03 [-0.68]	0.02 [0.51]	-0.06 [-1.29]	0.15 [3.10]	0.29 [3.66]
α_{FF4}	-0.14 [-2.19]	-0.02 [-0.31]	0.02 [0.42]	-0.05 [-1.10]	0.15 [2.97]	0.29 [3.55]
α_{FF5}	-0.22 [-3.67]	-0.06 [-1.12]	-0.05 [-1.23]	-0.18 [-3.94]	0.03 [0.73]	0.26 [3.22]
α_{FF6}	-0.22 [-3.55]	-0.04 [-0.79]	-0.05 [-1.15]	-0.16 [-3.51]	0.04 [0.89]	0.26 [3.22]
Panel B: Fama and French (2018) 6-factor model loadings for PRS-sorted portfolios						
β_{MKT}	0.99 [67.40]	1.00 [82.92]	0.99 [93.70]	1.00 [90.60]	0.97 [85.24]	-0.02 [-1.09]
β_{SMB}	-0.00 [-0.04]	-0.03 [-1.46]	-0.06 [-4.23]	-0.10 [-6.54]	-0.06 [-3.54]	-0.06 [-2.05]
β_{HML}	0.11 [3.93]	0.13 [5.58]	0.03 [1.62]	0.14 [6.50]	0.07 [3.41]	-0.04 [-0.98]
β_{RMW}	0.27 [9.29]	0.09 [3.90]	0.18 [8.67]	0.21 [9.67]	0.21 [9.30]	-0.06 [-1.59]
β_{CMA}	-0.05 [-1.11]	-0.04 [-1.10]	0.07 [2.41]	0.25 [7.97]	0.24 [7.52]	0.29 [5.26]
β_{UMD}	-0.01 [-0.39]	-0.02 [-1.92]	-0.00 [-0.40]	-0.03 [-2.30]	-0.01 [-0.98]	-0.01 [-0.28]
Panel C: Average number of firms (n) and market capitalization (me)						
n	405	354	317	352	391	
me (\$10 ⁶)	1150	1200	1690	1678	1956	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the PRS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.28 [3.51]	0.32 [4.01]	0.29 [3.66]	0.29 [3.55]	0.26 [3.22]	0.26 [3.22]
Quintile	NYSE	EW	0.33 [6.03]	0.39 [7.25]	0.33 [6.55]	0.28 [5.45]	0.24 [4.87]	0.20 [4.08]
Quintile	Name	VW	0.28 [3.38]	0.31 [3.82]	0.28 [3.43]	0.28 [3.30]	0.25 [3.04]	0.25 [3.01]
Quintile	Cap	VW	0.23 [3.03]	0.27 [3.48]	0.26 [3.37]	0.26 [3.24]	0.23 [3.02]	0.24 [2.99]
Decile	NYSE	VW	0.35 [3.81]	0.38 [4.14]	0.32 [3.49]	0.29 [3.17]	0.32 [3.47]	0.30 [3.24]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.24 [3.06]	0.28 [3.53]	0.25 [3.16]	0.25 [3.12]	0.23 [2.94]	0.23 [2.94]
Quintile	NYSE	EW	0.19 [3.31]	0.26 [4.60]	0.21 [3.84]	0.18 [3.33]	0.12 [2.37]	0.11 [2.05]
Quintile	Name	VW	0.24 [2.94]	0.27 [3.31]	0.24 [2.91]	0.23 [2.86]	0.22 [2.73]	0.22 [2.72]
Quintile	Cap	VW	0.20 [2.59]	0.23 [3.03]	0.22 [2.86]	0.22 [2.81]	0.21 [2.71]	0.21 [2.71]
Decile	NYSE	VW	0.31 [3.34]	0.33 [3.59]	0.27 [2.99]	0.26 [2.83]	0.27 [3.03]	0.27 [2.94]

Table 3: Conditional sort on size and PRS

This table presents results for conditional double sorts on size and PRS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on PRS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high PRS and short stocks with low PRS .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	PRS Quintiles					PRS Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.63 [2.75]	0.82 [3.61]	0.88 [3.19]	0.93 [4.36]	0.95 [4.55]	0.35 [3.80]	0.40 [4.31]	0.35 [3.90]	0.32 [3.51]	0.25 [2.72]	0.23 [2.48]
	(2)	0.65 [3.00]	0.78 [3.69]	0.83 [3.99]	0.97 [4.61]	0.85 [4.21]	0.20 [2.20]	0.24 [2.74]	0.19 [2.15]	0.20 [2.25]	0.13 [1.44]	0.15 [1.62]
	(3)	0.64 [3.31]	0.65 [3.26]	0.79 [4.04]	0.75 [3.98]	0.93 [4.92]	0.29 [3.41]	0.31 [3.69]	0.26 [3.08]	0.24 [2.83]	0.21 [2.52]	0.21 [2.38]
	(4)	0.52 [2.75]	0.66 [3.56]	0.74 [3.93]	0.66 [3.71]	0.81 [4.54]	0.29 [3.85]	0.33 [4.44]	0.28 [3.82]	0.29 [3.88]	0.19 [2.64]	0.21 [2.85]
	(5)	0.43 [2.56]	0.53 [3.21]	0.49 [3.04]	0.50 [3.09]	0.70 [4.50]	0.27 [2.93]	0.30 [3.29]	0.28 [3.06]	0.28 [2.94]	0.27 [2.90]	0.27 [2.86]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	PRS Quintiles					PRS Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	151	151	151	151	148	14	14	15	12	12	
	(2)	63	62	63	63	62	28	28	28	28	27	
	(3)	53	53	52	52	52	56	55	57	57	57	
	(4)	49	49	49	49	49	134	137	140	139	142	
(5)	52	52	52	52	52	1048	1249	1395	1249	1554		

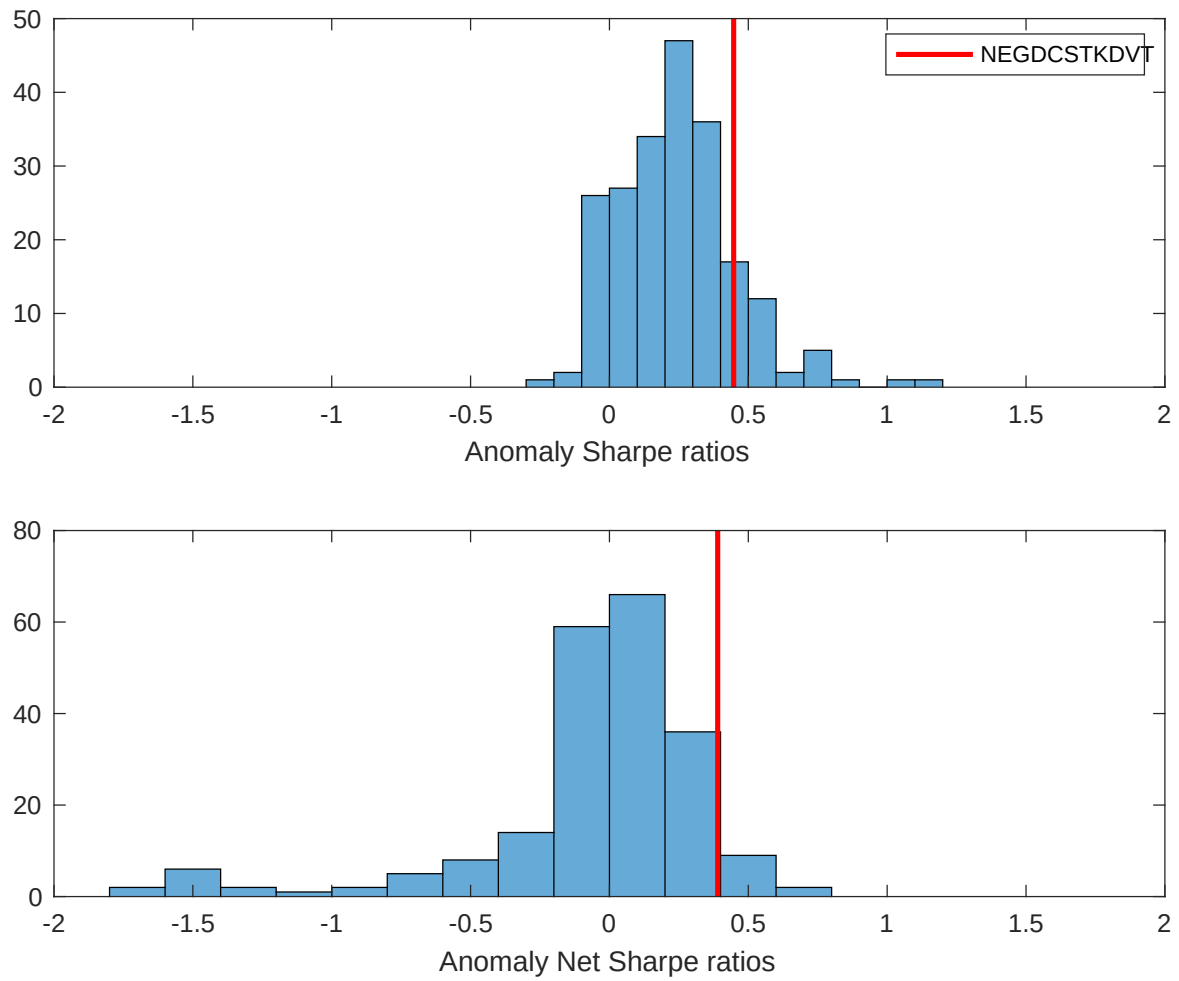


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the PRS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

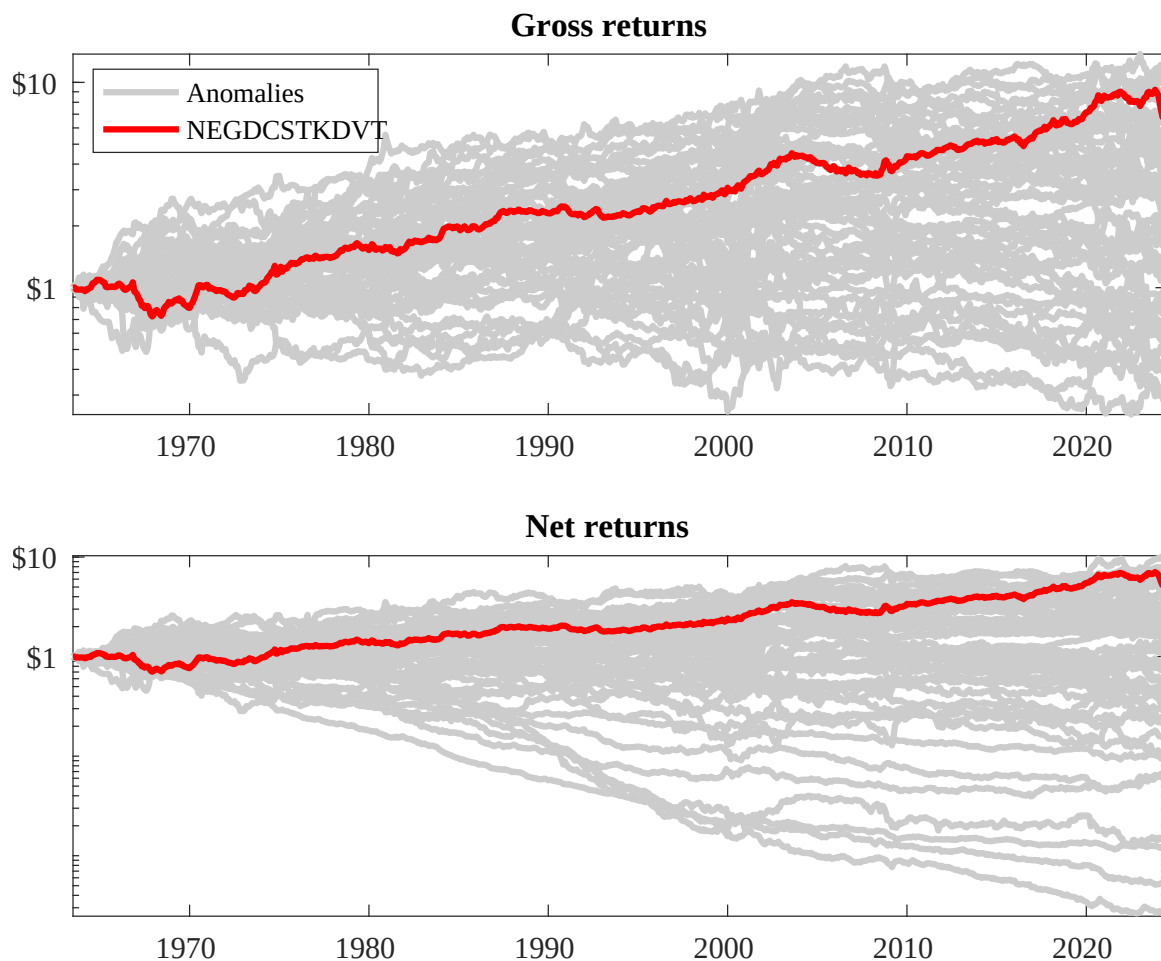
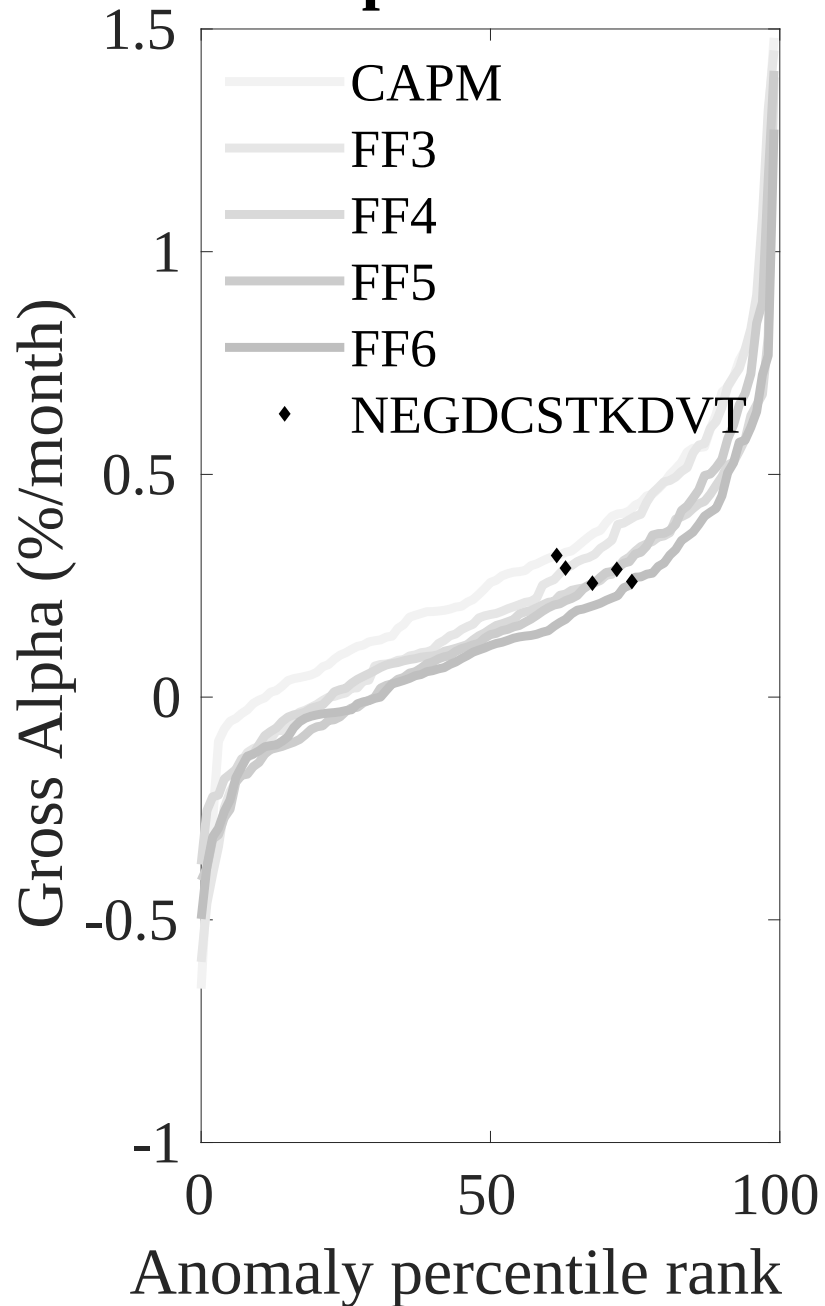


Figure 3: Dollar invested.

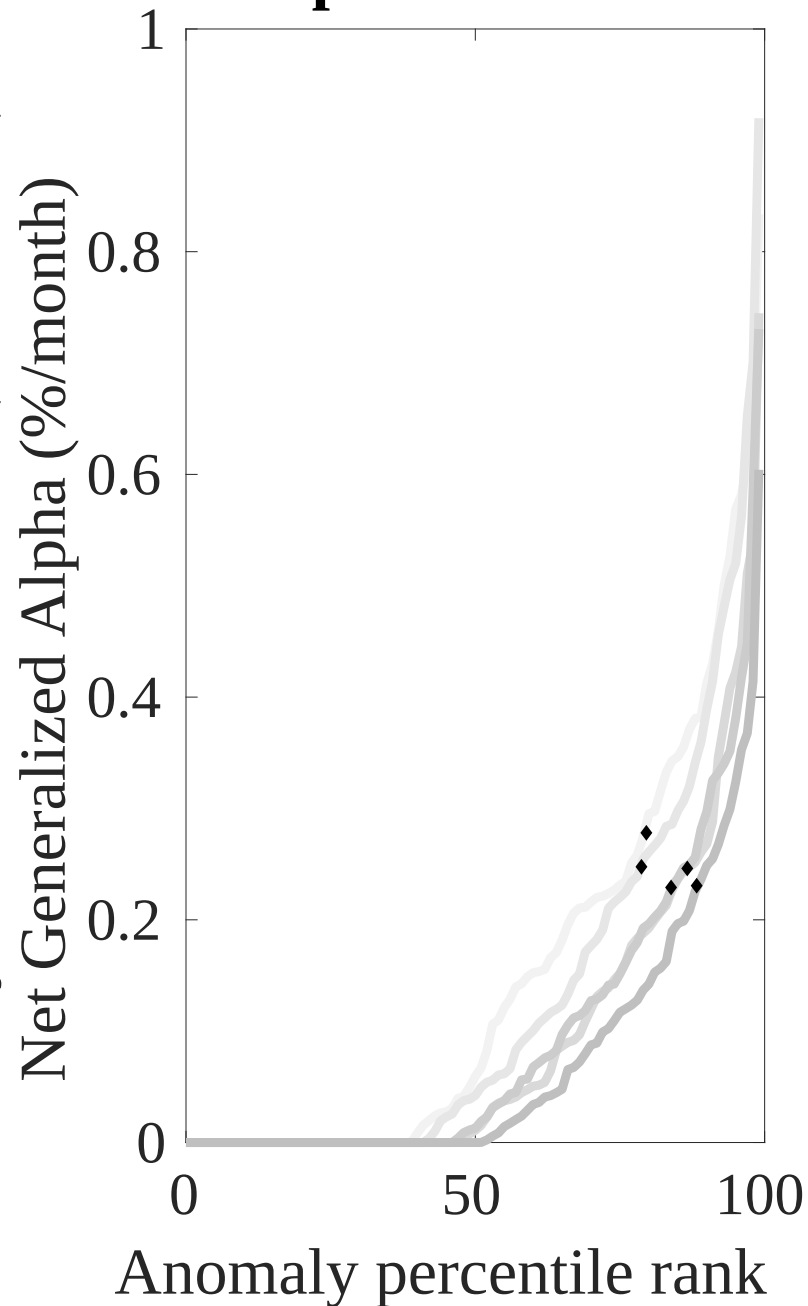
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the PRS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



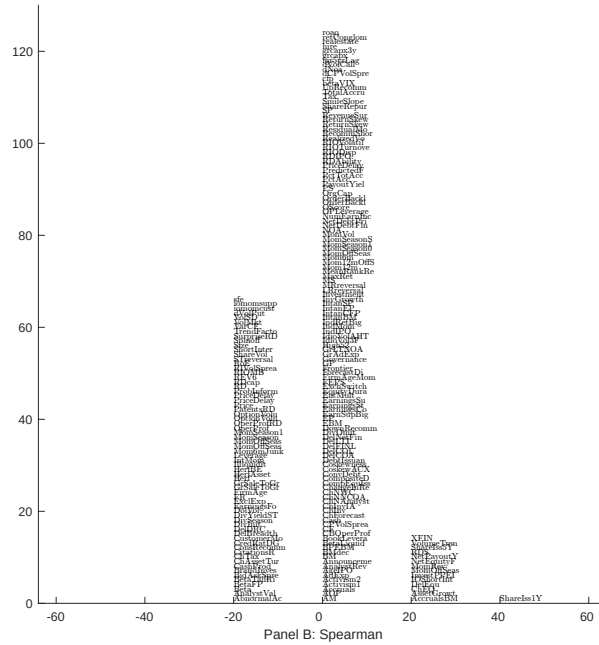
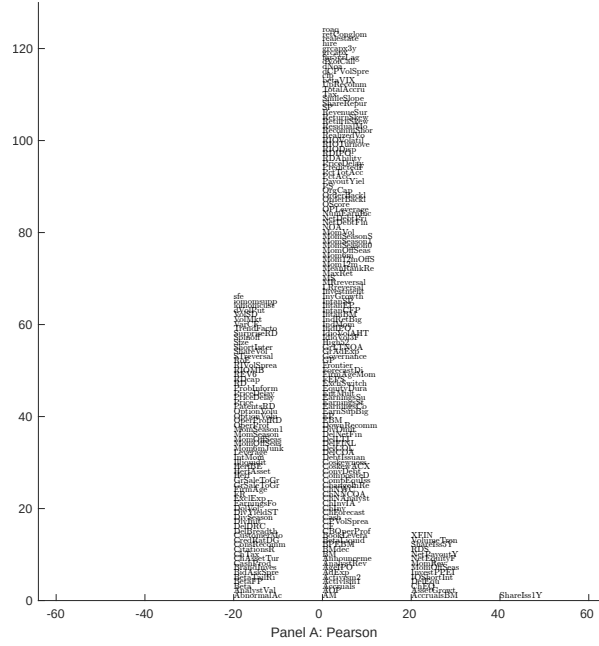


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 205 filtered anomaly signals with PRS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

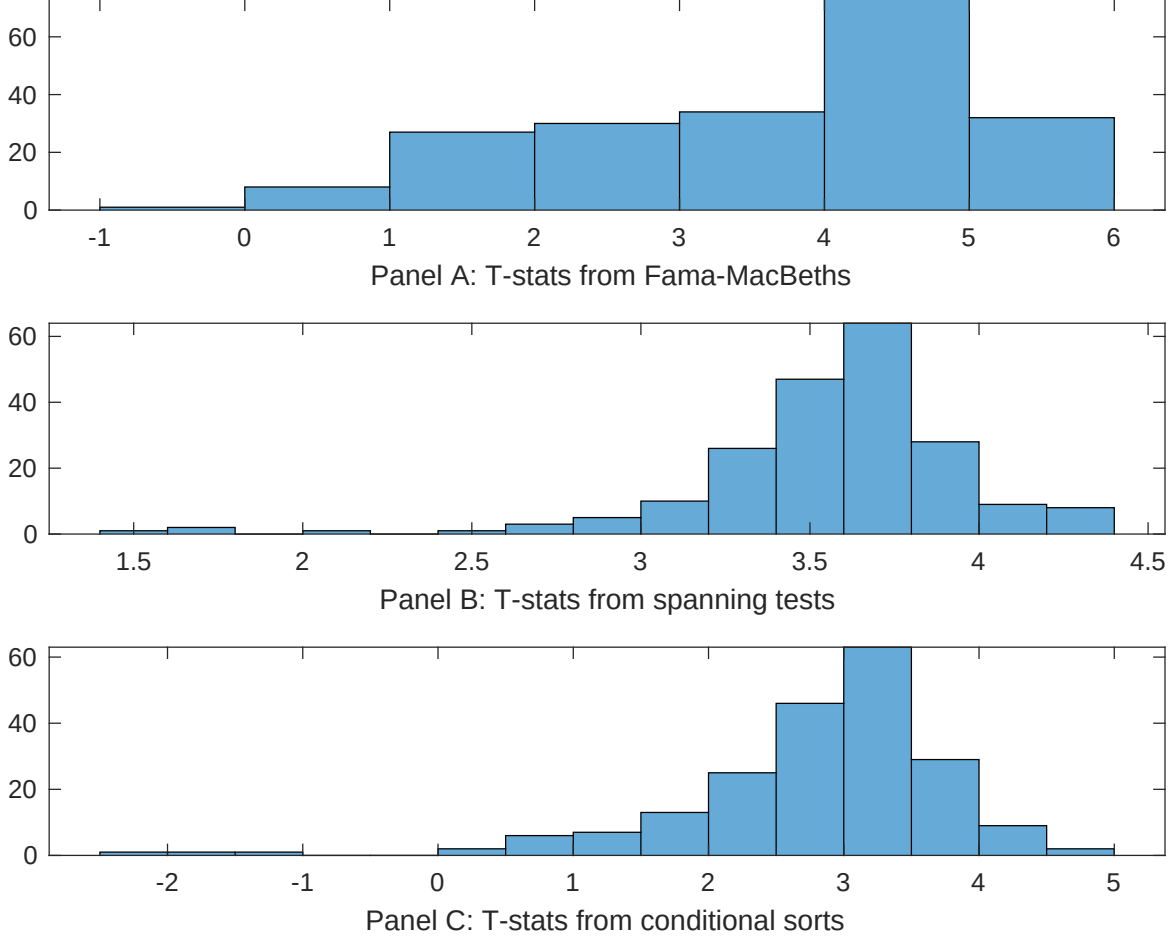


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of PRS conditioning on each of the 205 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PRS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PRS} PRS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 205 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PRS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 205 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 205 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on PRS. Stocks are finally grouped into five PRS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted PRS trading strategies conditioned on each of the 205 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on PRS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{PRS} PRS_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Growth in book equity, Net Payout Yield, Share issuance (1 year), Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.15 [6.41]	0.11 [5.53]	0.12 [6.45]	0.12 [6.65]	0.12 [6.28]	0.12 [6.66]	0.12 [3.87]
PRS	0.48 [3.24]	0.42 [1.74]	0.49 [3.43]	0.39 [2.60]	0.56 [3.68]	0.54 [3.56]	0.25 [1.03]
Anomaly 1	0.27 [2.10]						-0.22 [-0.09]
Anomaly 2		0.23 [2.06]					0.11 [1.00]
Anomaly 3			0.13 [5.51]				0.45 [1.49]
Anomaly 4				0.24 [3.98]			0.23 [3.71]
Anomaly 5					0.10 [2.50]		-0.11 [-0.17]
Anomaly 6						0.70 [5.48]	0.55 [4.28]
# months	720	715	715	715	720	720	715
$\bar{R}^2(\%)$	1	1	1	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the PRS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{PRS} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Growth in book equity, Net Payout Yield, Share issuance (1 year), Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.24 [3.08]	0.29 [3.67]	0.25 [3.16]	0.18 [2.32]	0.25 [3.14]	0.25 [3.09]	0.18 [2.44]
Anomaly 1	23.29 [5.44]						37.22 [6.23]
Anomaly 2		16.53 [5.45]					8.35 [2.55]
Anomaly 3			24.31 [6.16]				7.39 [1.57]
Anomaly 4				31.63 [7.70]			20.53 [4.41]
Anomaly 5					2.36 [0.59]		-18.08 [-3.29]
Anomaly 6						-14.25 [-2.99]	-24.52 [-4.77]
mkt	-1.21 [-0.64]	0.83 [0.43]	-1.13 [-0.61]	-0.33 [-0.18]	-2.01 [-1.05]	-1.99 [-1.04]	2.07 [1.13]
smb	-5.37 [-1.97]	-1.87 [-0.68]	-3.61 [-1.34]	-2.41 [-0.90]	-4.62 [-1.66]	-3.45 [-1.24]	-0.37 [-0.14]
hml	-5.24 [-1.42]	-8.36 [-2.19]	-2.53 [-0.70]	0.54 [0.15]	-2.66 [-0.71]	-1.83 [-0.49]	-5.39 [-1.48]
rmw	-4.95 [-1.34]	-15.22 [-3.75]	-18.75 [-4.45]	-16.95 [-4.38]	-5.83 [-1.55]	-5.84 [-1.57]	-21.61 [-4.97]
cma	5.51 [0.81]	13.61 [2.32]	12.41 [2.14]	12.25 [2.21]	25.70 [3.79]	45.28 [5.72]	18.20 [2.32]
umd	-0.77 [-0.41]	0.99 [0.53]	-1.43 [-0.77]	-2.39 [-1.30]	-0.28 [-0.15]	-0.76 [-0.40]	-3.18 [-1.77]
# months	732	728	728	728	732	732	728
$\bar{R}^2(\%)$	11	11	12	14	7	8	21

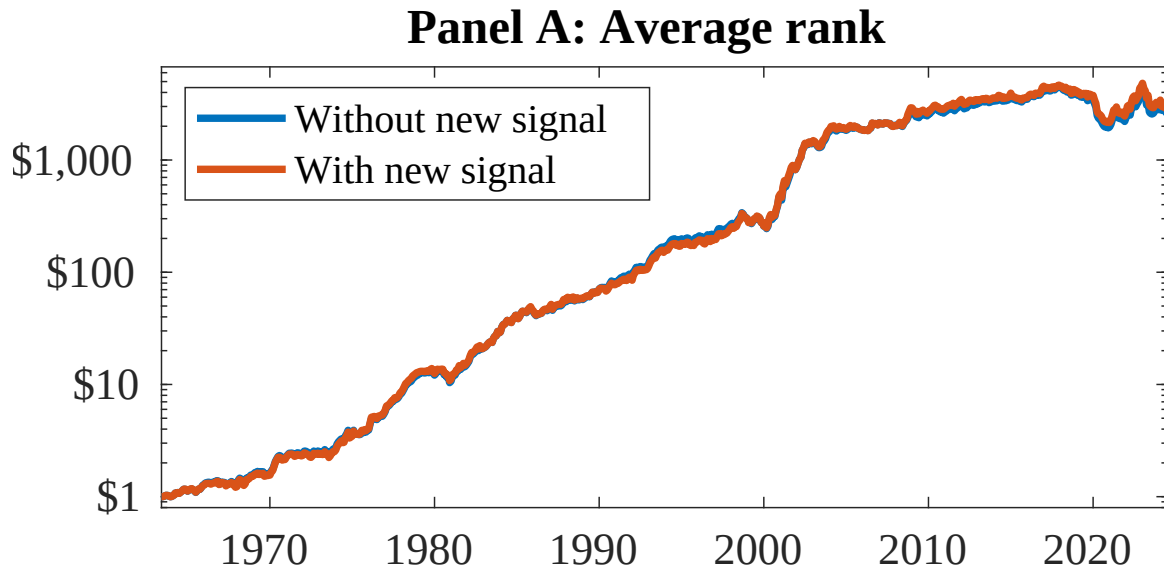


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as PRS. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

- Barberis, N. and Shleifer, A. (2003). Style investing. *Journal of Financial Economics*, 68(2):161–199.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- DellaVigna, S. and Pollet, J. M. (2009). Investor inattention and friday earnings announcements. *Journal of Finance*, 64(2):709–749.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2001). Disappearing dividends: changing firm characteristics or lower propensity to pay? *Journal of Financial Economics*, 60(1):3–43.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Grullon, G. and Michaely, R. (2002). Dividends, share repurchases, and the substitution hypothesis. *Journal of Finance*, 57(4):1649–1684.

- Hirshleifer, D. and Teoh, S. H. (2003). Limited attention, information disclosure, and financial reporting. *Journal of Accounting and Economics*, 36(1-3):337–386.
- Hong, H., Lim, T., and Stein, J. C. (2000). Bad news travels slowly: Size, analyst coverage, and the profitability of momentum strategies. *Journal of Finance*, 55(1):265–295.
- Hong, H. and Stein, J. C. (1999). A unified theory of underreaction, momentum trading, and overreaction in asset markets. *Journal of Finance*, 54(6):2143–2184.
- Hong, H. and Stein, J. C. (2007). Disagreement and the stock market. *Journal of Economic Perspectives*, 21(2):109–128.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Pástor, Ľ. and Veronesi, P. (2003). Stock valuation and learning about profitability. *Journal of Finance*, 58(5):1749–1789.